Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An autonomous Institution affiliated to Anna University)

Degree & Branch	BE Computer Science & Engineering	**Semester**	VI
Subject Code & Name	Machine Learning Laboratory		
Academic Year	2025–2026	**Batch**	2023–2027

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Experiment 9: Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

1. Aim and Objectives

To implement and analyze the performance of clustering algorithms on the Human Activity Recognition dataset:

- **Model A:** K-Means Clustering.
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise).
- **Model C:** Hierarchical Agglomerative Clustering (HAC).

Students must visualize and compare the clusters formed by each algorithm, report results, and analyze performance.

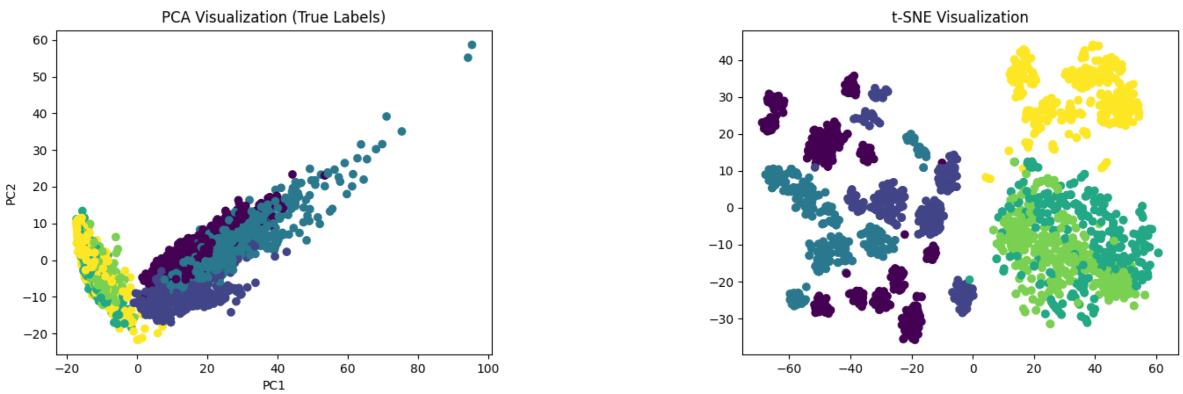
2. Dataset Description

Download: Human Activity Recognition Using Smartphones Dataset
- 30 volunteers (aged 19–48). - 6 activities: WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING. - Sensors: 3-axis accelerometer and gyroscope (50 Hz). - Preprocessed into windows of 2.56s (128 readings/window) with feature extraction from time and frequency domain.

3. Preprocessing Steps

- **Data Loading:** The dataset was loaded from text files into `pandas` DataFrames. Specifically, `X_train.txt`, `y_train.txt`, `X_test.txt`, and `y_test.txt` were used. `X_train` and `X_test` contain feature vectors, while `y_train` and `y_test` contain activity labels.
- **Data Inspection:** `X_train` contains 561 numerical (`float64`) features with no missing values.
- **Feature Visualization:** Histograms and boxplots of selected features (`X_train.iloc[:, :5]`) were plotted to analyze distributions and detect outliers.

- **Standardization:** Features were scaled using `StandardScaler` to achieve zero mean and unit variance, which is essential for distance-based clustering.
- **Dimensionality Reduction:**
  - PCA (`n_components=2`) was applied to obtain `X_pca` for visualization.
  - t-SNE (`n_components=2`, `perplexity=30`) was applied on a subset (2000 samples) to obtain `X_tsne`.



(a) Scatter plots of clusters after t-SNE

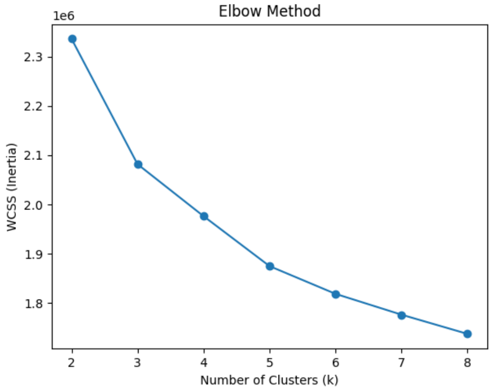
(b) Scatter plots of clusters after PCA

Figure 1: Scatter plots of clusters after PCA and t-SNE

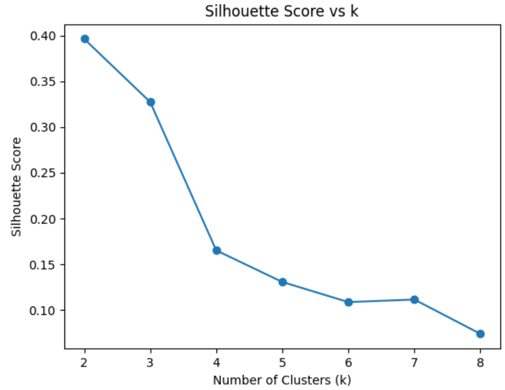
4. Clustering Implementation Details

4.1 K-Means Clustering

- **Objective:** Partition data into k clusters based on nearest mean.
- **Elbow Method:** WCSS was computed for $k = 2$ to 8. The elbow point was observed at $k = 6$.
- **Silhouette Score:** Highest at $k = 2$ (0.3965), but $k = 6$ was selected for better granularity.
- **Implementation:** `KMeans(n_clusters=6, random_state=42, n_init=10)`
- **Visualization:** PCA scatter plot colored by cluster labels.



(a) K Vs WCSS



(b) K Vs Silhouette score

Table 1: K-Means Elbow Method Results

Number of Clusters (k)	**WCSS (Inertia)**	**Silhouette Score**
2	2336223.75	0.3965
3	2081849.63	0.3274
4	1976452.64	0.1652
5	1875066.09	0.1306
6	1818790.07	0.1086
7	1776518.73	0.1114
8	1737555.44	0.0739

4.2 DBSCAN Clustering

- **Objective:** Identify dense regions and label sparse points as noise.
- **Parameter Selection:** Based on K-distance graph, $\epsilon = 20.0$, `min_samples = 10`.
- **Implementation:** `DBSCAN(eps=20.0, min_samples=10)`
- **Observation:** Initial $\epsilon = 8.0$ resulted in excessive noise.
- **Visualization:** PCA scatter plot showing clusters and noise.

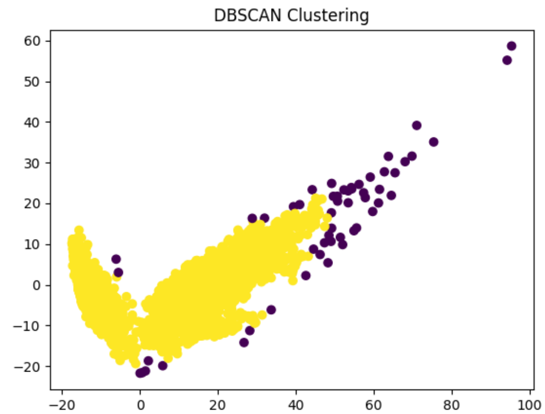
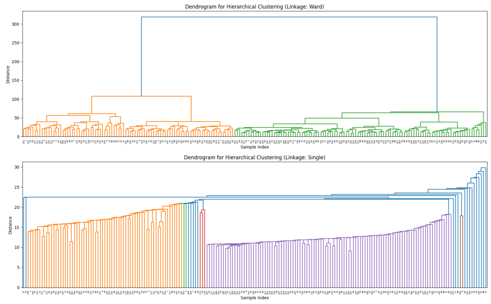


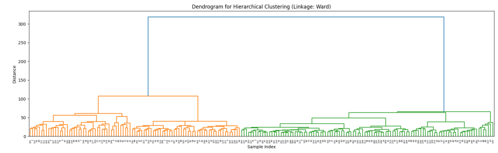
Figure 3: Scatter plot of DBSCAN Clustering

4.3 Agglomerative Clustering

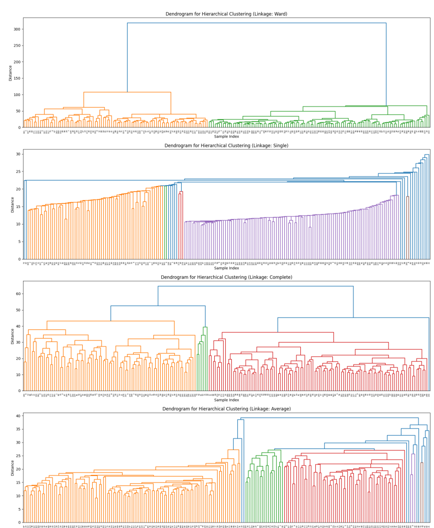
- **Objective:** Build hierarchical clusters by merging smaller clusters.
- **Implementation:** `AgglomerativeClustering(n_clusters=6, linkage='ward')`
- **Dendrogram:** Generated using a subset of 200 samples.
- **Visualization:** PCA scatter plot with cluster assignments.



(a) Single linkage



(b) Ward linkage



(c) Average linkage

(d) Complete linkage

Figure 4: Dendrogram of various linkages

5. Evaluation Metrics and Results

To evaluate clustering performance against true labels (`y_train`), the following metrics were used:

- **True Labels Preparation:** `y_train` was flattened to match cluster labels.
- **Adjusted Rand Index (ARI):** Measures similarity between predicted and true clusters.
- **Normalized Mutual Information (NMI):** Measures shared information between clusterings.

Performance Metrics Summary

Table 2: Clustering Performance Metrics

Clustering Method	ARI	NMI
K-Means	0.4197	0.5589
DBSCAN (refined)	0.0079	0.0378
Agglomerative Clustering	0.2794	0.4544

6. Observations and Comparative Analysis

K-Means (k=6)

- Achieved the highest ARI (0.4197) and NMI (0.5589).
- Produced well-separated clusters in PCA visualization.

DBSCAN

- Very low ARI (0.0079) and NMI (0.0378).
- Large portion of data classified as noise.

Agglomerative Clustering

- Moderate performance (ARI: 0.2794, NMI: 0.4544).
- Cluster boundaries less distinct than K-Means.

7. Conclusion

K-Means clustering with $k = 6$ proved to be the most effective method, achieving the highest agreement with true labels. DBSCAN struggled due to density sensitivity, while Agglomerative Clustering showed moderate performance.

8. Learning Outcomes

- Understood the importance of data preprocessing techniques such as standardization and dimensionality reduction.
- Learned how to apply clustering algorithms including K-Means, DBSCAN, and Agglomerative Clustering.
- Gained practical knowledge of selecting optimal parameters using methods like the Elbow Method and Silhouette Score.
- Developed skills in visualizing high-dimensional data using PCA and t-SNE.
- Learned how to evaluate clustering performance using metrics such as Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI).
- Analyzed and compared different clustering techniques to understand their strengths and limitations.
- Improved ability to interpret clustering results and relate them to real-world datasets.